Exact Optimal Stretch of Period-Three Temporal Cycles

Research Note No. 27 — Unrefereed Preprint v1.0

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15 October 2026

Abstract

We consider the optimization form of stretched temporal graph realization (STGR) introduced by Mertzios et al., restricted to an undirected cycle C_n and period $\Delta = 3$. With one edge-availability phase per period and fastest strict temporal paths, we derive an exact closed-form expression for the optimal multiplicative stretch: 1 for $n = 3, 4$, 3/2 for $n = 5$ and even $n \geq 6$, and 5/3 for odd $n \geq 7$. Explicit cyclic labelings provide the upper bounds; short local constraints on oriented walks give the lower bounds, including the exceptional seven-cycle. Two differently implemented exhaustive checks agree on $3 \leq n \leq 9$. The analytic proof and computational evidence have passed independent review. This is an unrefereed preprint; historical novelty and priority remain unestablished.

1 Model and statement

Let C_n have vertices $0, \ldots, n-1$ and edges $e_i = \{i, i+1\}$ (indices modulo n). Assign each edge a phase $\lambda_i \in \mathbb{Z}_3$. Edge e_i is usable at all integer times congruent to λ_i modulo 3, with temporal path traversal times strictly increasing. As in STGR [1], the duration of a temporal path with edge phases $a_1, \ldots, a_d$ is the last traversal time minus the first plus one; this equals

$$T(a_1, \ldots, a_d) = 1 + \sum_{j=1}^{d-1} w(a_j, a_{j+1}), \qquad w(a,b) = \begin{cases} (b-a) \bmod 3, & a \not\equiv b \pmod 3, \\ 3, & a \equiv b \pmod 3. \end{cases} \tag{1}$$

For an ordered vertex pair (u, v), let $T_\lambda(u, v)$ be the smallest duration of either of the two simple directed arcs of C_n, and let $d(u, v) = \min\{(v-u) \bmod n, (u-v) \bmod n\}$. Define

$$\alpha^*(C_n, 3) = \min_{\lambda \in \mathbb{Z}_3^n} \max_{u \neq v} \frac{T_\lambda(u, v)}{d(u, v)}.$$

The objective and the temporal-graph model are established in the STGR literature; only the cycle/period-three exact classification below is under consideration here. The model follows the strict, earliest-possible traversal convention and the ordered-pair definition of stretch in Mertzios et al. (MFCS 2025, Section 2); phases $0, 1, 2$ are a relabeling of their $1, 2, 3$.

Theorem 1 (Exact classification)**.** *For every $n \geq 3$,*

$$\boxed{\alpha^*(C_n, 3) = \begin{cases} 1, & n = 3, 4, \\ \frac{3}{2}, & n = 5 \text{ or even } n \geq 6, \\ \frac{5}{3}, & \text{odd } n \geq 7. \end{cases}}$$

2 Upper bounds: explicit labels

We write phase words $\lambda_0 \cdots \lambda_{n-1}$ in cyclic edge order.

Lemma 1. *The values in Theorem 1 are attainable as upper bounds.*

Proof. For $n = 3$, all distinct vertex pairs are adjacent, so the value is 1. For $n = 4$, use the cyclic word 0101. For every opposite pair at least one of its two 2-edge arcs has duration 2, so the value is also 1.

For $n = 5$, choose 02101. All adjacent edge phases differ. The diameter is 2, and every 2-edge arc has duration at most 3, so the stretch is at most 3/2.

For even $n \geq 6$, take $\lambda = (01)^{n/2}$. Traversing either direction along any simple arc of d edges, the successive waits alternate between 1 and 2. Thus

$$T \leq d + \left\lceil \frac{d-1}{2} \right\rceil \leq \frac{3d}{2}.$$

For $n = 7$, take 0210101. Adjacent phases are different, and every simple arc of $d \leq 3$ edges has duration at most $1 + 2(d-1)$. This yields ratios $1, 3/2, 5/3$ for $d = 1, 2, 3$, respectively.

Finally, let $n \geq 9$ be odd, put $k = (n-7)/2$, and use

$$\lambda = 0202121(01)^k.$$

Its cyclic differences $g_i = (\lambda_{i+1} - \lambda_i) \bmod 3$ lie in $\{1, 2\}$ and no three consecutive differences are equal. Indeed, the initial cyclic difference pattern (including the junction into the repeated 01 part) is $2, 1, 2, 2, 1, 2, 2, 1, 2$, followed, when $k > 1$, by an alternating $1, 2$ tail; the wrap-around also preserves the property. Reversing an arc replaces every wait g by $3 - g$, so the same no-three-equal property holds for directed waits in either orientation. Any directed arc of d edges has $m = d - 1$ waits, at most $\lceil 2m/3 \rceil$ of which can equal 2, and hence

$$T \leq d + \left\lceil \frac{2(d-1)}{3} \right\rceil \leq \frac{5d}{3}.$$

Taking a shortest static arc between each pair proves the upper bound. □

3 Lower bounds

Lemma 2. *For all $n \geq 5$, $\alpha^*(C_n, 3) \geq 3/2$.*

Proof. Consider two consecutive edges with phases a, b. Their 2-edge arc in one direction has duration $1 + w(a, b)$, and the reverse 2-edge arc has duration $1 + w(b, a)$. The two waits cannot both equal 1: if $a = b$ they are both 3, otherwise their sum is 3. At least one orientation therefore has 2-edge duration at least 3. For $n \geq 5$, the alternative simple arc has $n - 2 \geq 3$ edges and duration at least 3. Thus some ordered pair at static distance 2 has fastest duration at least 3, and stretch at least 3/2. □

Lemma 3. *For odd $n \geq 9$, $\alpha^*(C_n, 3) \geq 5/3$.*

Proof. Assume a labeling has stretch strictly below 5/3. For each oriented 2-edge arc, the alternative has $n - 2 \geq 7$ edges, so the 2-edge arc itself must have duration at most 3. Consequently consecutive edge phases differ; write $g_i = (\lambda_{i+1} - \lambda_i) \bmod 3 \in \{1, 2\}$.

For each oriented 3-edge arc, the alternative has $n - 3 \geq 6$ edges, so the 3-edge arc itself must have duration at most 4, since $3(5/3) = 5$ and durations are integral. Taking the clockwise arc e_i, e_{i+1}, e_{i+2} gives $1 + g_i + g_{i+1} \leq 4$. Taking the reverse orientation of the same three edges gives $1 + (3 - g_i) + (3 - g_{i+1}) \leq 4$. Therefore $g_i + g_{i+1} = 3$ for every i, i.e. g_i must alternate $1, 2, 1, 2, \ldots$ around the cycle. That is impossible on odd n. □

Lemma 4. $\alpha^*(C_7, 3) \geq 5/3$.

Proof. Suppose a labeling has stretch below 5/3. Each 2-edge arc must have duration at most 3, because its alternative has 5 edges. Hence $g_i \in \{1, 2\}$ for all i. Since $\sum_i g_i \equiv 0 \pmod 3$, the number of 2's among the seven gaps is either 2 or 5. Reversing the orientation of the entire cycle interchanges 1 and 2; assume without loss of generality that there are five 2's.

Some adjacent gaps are both 2 (otherwise there would be at most three 2's). After rotating indices, let $g_i = g_{i+1} = 2$. The clockwise 3-edge arc on e_i, e_{i+1}, e_{i+2} takes duration 5; to obtain stretch below 5/3 for its ordered endpoints, the complementary 4-edge reverse arc must take duration at most 4. Its waits are $3 - g_{i+5}$, $3 - g_{i+4}$, and $3 - g_{i+3}$, so this requires $g_{i+3} = g_{i+4} = g_{i+5} = 2$. These are all five 2's, leaving $g_{i+2} = g_{i+6} = 1$.

Now the clockwise 3-edge arc with waits $g_{i+3} = g_{i+4} = 2$ also takes duration 5. Its complementary reverse 4-edge arc has one wait $3 - g_{i+6} = 2$, so its duration is at least 5 as well. This ordered pair has static distance 3 and fastest duration at least 5, contradicting stretch below 5/3. □

Proof of Theorem 1. The universal bound $\alpha^* \geq 1$ is immediate because no path has fewer edge traversals than static distance. Combine Lemma 1 with Lemmas 2, 3, and 4. □

4 Reproducibility and scope

We supply an exact rational enumeration of all 3^{n-1} phase-normalized labelings for $3 \le n \le 9$ and a separate clock-time simulation implementation of the same enumeration. Their optimal values are respectively $1, 1, 3/2, 3/2, 5/3, 3/2, 5/3$ for $n = 3, \ldots, 9$. The clock-time implementation also rules out all 42 phase-consistent $\{1, 2\}$-gap sequences on C_7 under a hypothetical stretch $< 5/3$ by checking both directions of the 3-edge arcs. Infinite-family witnesses are checked computationally through $n = 101$. These finite tests *support* but do not replace the analytic proofs.

Novelty limitation. The STGR model and objective are prior art [1], with subsequent work on possible stretch values and period two [2]. To the best of our knowledge, the specific exact classification for $(C_n, \Delta = 3)$ was not identified in the literature examined. However, we do not claim priority; a complete historical assessment remains open. The reported independent proof and reproducibility review passed on 15 October 2026, with eight original payload hashes matching. The supplementary reviewer tests through $n = 14$ were reported independently and are not included as raw computational evidence here. This preprint has not undergone journal or conference peer review. For earlier or overlapping results, contact yonezaemon1+research@gmail.com.

References

[1] G. B. Mertzios, H. Molter, N. Morawietz, and P. G. Spirakis, *Temporal Graph Realization with Bounded Stretch*, MFCS 2025, LIPIcs 345, article 75. https://doi.org/10.4230/LIPIcs.MFCS.2025.75. Journal version: JCSS 160 (2026), 103814.

[2] J. Meusel, N. Morawietz, M. Müller-Hannemann, and K. Reinhardt, *Brief Announcement: Revisiting the Realizability of Periodic Temporal Graphs with Bounded Stretch*, SAND 2026, LIPIcs 373, article 21. https://doi.org/10.4230/LIPIcs.SAND.2026.21.

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